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Abbreviation

CNN = Convolutional Neural Network
LSTM = Long Short-Term Memory
BiLSTM = Bidirectional Long Short-Term Memory
ReLU = Rectified Linear Unit
RMSE = Root Mean Square Error
MAE = Mean Absolute Error
MSE = Mean Square Error
R2 = Coefficient of Determination
ANFIS = Adaptive Neuro-Fuzzy Inference System
ICA = Independent Component Analysis

Proposed Title

A Hybrid Learning for Detecting and Classifying Transmission Line Faults Using ChronoGrid FusionNet Model

Proposed Title Description

- **A:** Represents a unified and structured deep learning framework designed for power system protection.
- **Hybrid:** Indicates integration of **One-Dimensional Convolutional Neural Network**, **Bidirectional Long Short-Term Memory Network**, and **Self-Attention Mechanism**.
- **Learning:** Refers to supervised deep representation learning from raw multi-channel voltage and current time-series data.
- **Detecting:** Emphasizes binary discrimination between fault and non-fault operating states.
- **and:** Connects the dual objectives of detection and multi-class classification within a single architecture.
- **Classifying:** Focuses on categorizing different asymmetrical fault types with high separability.
- **Transmission Line Faults:** Specifies the application domain within high-voltage interconnected smart grids.
- **Using:** Highlights the methodological approach through computational intelligence techniques.
- **ChronoGrid FusionNet Model:** Our proposed hybrid architecture combining convolutional feature extraction, sequential temporal modeling, and adaptive attention weighting.

Proposed Abstract

Transmission line faults pose serious threats to the stability and reliability of modern interconnected power systems. In systems such as the IEEE 9-bus network, asymmetrical faults including single-line-to-ground, double-line, and double-line-to-ground faults generate nonlinear transient disturbances in voltage and current waveforms. The base study introduces a time-series-based **Convolutional Neural Network** model that directly processes raw six-channel post-fault signals without handcrafted feature engineering. By employing one-dimensional convolutional layers with **Rectified Linear Unit** activation functions, the architecture extracts localized temporal distortions and transient waveform characteristics. Experimental validation demonstrates near-zero **Mean Square Error** for fault detection and satisfactory performance for multi-class classification within simulated IEEE 9-bus conditions.

However, the existing system reveals several limitations. The shallow **Convolutional Neural Network** primarily captures localized signal variations and lacks the ability to model long-range temporal dependencies inherent in sequential voltage-current patterns. Similar asymmetrical faults exhibit overlapping waveform signatures, causing misclassification under regression-based output layers. The model evaluation is restricted to simulated datasets, which limits generalization to real-world noisy environments and larger bus systems. Moreover, the absence of memory-based sequential modeling such as **Long Short-Term Memory Network** and the lack of adaptive attention mechanisms reduce contextual understanding of transient intervals.

To overcome these shortcomings, we propose the ChronoGrid FusionNet Model, a hybrid architecture integrating **One-Dimensional Convolutional Neural Network**, **Bidirectional Long Short-Term Memory Network**, and **Self-Attention Mechanism**. The convolutional layers perform multi-scale temporal feature extraction, while the bidirectional memory network captures forward and backward dependencies across time steps. The attention mechanism assigns adaptive weights to critical transient segments, enhancing discrimination between closely related faults. Compared to the baseline **Convolutional Neural Network**, the proposed hybrid framework improves classification accuracy, enhances robustness under noisy conditions, and maintains computational efficiency for real-time smart grid protection.

<!-- <h1 id="introduction">Introduction
-->

Real Time Use Cases

In modern substations, intelligent electronic devices require rapid and precise fault identification to isolate damaged transmission segments. The proposed ChronoGrid FusionNet Model can be deployed within digital protective relays to analyze streaming voltage and current time-series data. When transient disturbances occur, the hybrid system detects fault presence and accurately classifies the fault type within milliseconds. This reduces relay operation delay, minimizes equipment damage, and enhances transmission reliability in interconnected smart grids.

Another important application is in wide-area monitoring systems supported by synchronized phasor measurement units. These systems generate high-resolution multi-bus time-series datasets. The proposed hybrid architecture processes these synchronized measurements to support adaptive protection and predictive maintenance strategies. By leveraging **Bidirectional Long Short-Term Memory Network** and **Self-Attention Mechanism**, the system improves fault discrimination in renewable-integrated grids where transient behaviors are highly dynamic. This enhances grid resilience and supports self-healing operations.

Existing Algorithms

- Extreme Learning Machine
- Convolutional Neural Network with Continuous Wavelet Transform
- Convolutional Neural Network–Long Short-Term Memory Hybrid Model

Existing System Disadvantages

- **Insufficient Temporal Dependency Modeling:** The standalone **Convolutional Neural Network** architecture focuses on localized convolutional filtering but does not effectively capture long-term sequential correlations in voltage and current time-series data, limiting classification precision for similar fault types.
- **Regression-Based Classification Ambiguity:** Using regression output layers instead of probabilistic classification functions causes overlapping decision boundaries between asymmetrical fault categories, resulting in reduced separability.
- **Limited Dataset Diversity:** Evaluation on only simulated IEEE 9-bus systems restricts robustness validation under real-world grid conditions, noise disturbances, and topology variations.
- **Shallow Feature Representation:** The relatively simple architecture restricts hierarchical feature abstraction, making it difficult to differentiate subtle waveform distortions in complex transient scenarios.
- **No Adaptive Attention Mechanism:** The absence of attention-based weighting prevents dynamic prioritization of critical transient intervals during fault events, reducing contextual learning capacity.

Literature Survey

Literature Survey 1

A Two-Stage Framework for Fault and Cyber Attack Detection in Line Current Differential Relays of Power Transmission Network

Authors: Anfaj Islam, Duy-Hung Ha, Trung-Thang Nguyen, Van-Van Huynh, Van Khang Huynh

Published Date: 14-Nov-2025

DOI: 10.1109/ACCESS.2025.3632972

Literature

The increasing integration of digital platforms in power systems has enhanced protection capabilities but also introduced new vulnerabilities, particularly in critical components like Line Current Differential Relays (LCDRs). LCDRs offer fast, selective, and sensitive protection for high-voltage transmission lines, but their reliance on synchronized current measurements, communicated over potentially insecure networks, exposes them to cyber threats. Attackers can exploit this dependency through sophisticated attacks, namely False Data Injection Attacks (FDIA) and Time Synchronization Attacks (TSA), manipulating relay decisions and leading to unnecessary tripping, load shedding, and system instability. To address this challenge, this study proposes a two-stage detection and classification framework to distinguish between grid faults and cyber-attacks. The first stage employs an unsupervised Long Short-Term Memory autoencoder, trained solely on normal operational data, to detect anomalous deviations without requiring labeled attack data. Once an anomaly is detected, the second stage, which is based on a supervised Random Forest classifier, categorizes the event as either a physical fault or a specific type of cyberattack. This research framework is implemented and validated on the IEEE 14-bus system using the OPAL-RT HYPERSIM platform. Results demonstrate that this research framework can accurately detect and classify both faults and cyber-attacks while preserving the essential protective functions of the LCDR.

Literature Survey 2

Improved Relay Algorithm for Detection and Classification of Transmission Line Faults in Monopolar HVDC Transmission System Using Signum Function of Transient Energy

Authors: Soma Deb, Suman Lata, Vikas Singh Bhadoria, Shubham Tiwari, Theodoros I. Maris, Vasiliki Vita, Georgios Fotis

Published Date: 19-Jan-2024

DOI: 10.1109/ACCESS.2024.3356012

Literature

Based on the signum function (sign of magnitudes) of transient energy, this paper proposes a fault classification approach and algorithm for a monopolar HVDC system. The analytical study here shows that the signum function of variation in transient energy has a zero value individually at both ends on no-fault conditions. Moreover, the summation of the signum functions computed for internal DC faults is zero, whereas the same has a non-zero value for external AC faults. The performance of this research algorithm has been extensively evaluated by simulating faults on a CIGRE benchmark system for HVDC monopolar configuration using EMTDC/PSCAD software. Three locations of DC line fault, and AC fault at the two ends of the system have been considered for this evaluation. Five fault resistance values (0, 10, 100, 1000, and 2000 ohms) have been simulated for each fault location. The results conform with the theoretical analysis, and the fault classification by the algorithm is 100% accurate. The time taken to detect and classify a DC fault at the mid-point of the 800-km line is 1.5 ms, and that for line-end faults on the DC line is 3 ms for all values of fault resistance. These results show a marked improvement over those reported earlier in the literature using other techniques. A comparison table is given in the last section to corroborate it.

Literature Survey 3

Single-ended Fault Detection Scheme Using Support Vector Machine for Multi-terminal Direct Current Systems Based on Modular Multilevel Converter

Authors:

Published Date: 04-Feb-2022

DOI: 10.35833/MPCE.2021.000404

Literature

This paper proposes a single-ended fault detection scheme for long transmission lines using support vector machine (SVM) for multi-terminal direct current systems based on modular multilevel converter (MMC-MTDC). The scheme overcomes existing detection difficulties in the protection of long transmission lines resulting from high grounding resistance and attenuation, and also avoids the sophisticated process of threshold value selection. The high-frequency components in the measured voltage extracted by a wavelet transform and the amplitude of the zero-mode set of the positive-sequence voltage arethe inputs to a trained SVM. The output of the SVM determines the fault type. A model of a four-terminal DC power grid with overhead transmission lines is built in PSCAD/EMTDC. Simulation results of EMTDC confirm that this research scheme achieves 100% accuracy in detecting short-circuit faults with high resistance on long transmission lines. This research scheme eliminates mal-operation of DC circuit breakers when faced with power order changes or AC-side faults. Its robustness and time delay are also assessed and shown to have no perceptible effect on the speed and accuracy of the detection scheme, thus ensuring its reliability and stability.

Literature Survey 4

A simple decision tree-based disturbance monitoring system for VSC-based HVDC transmission link integrating a DFIG wind farm

Authors:

Published Date: 01-Apr-2022

DOI: 10.1186/s41601-022-00247-w

Literature

Fault detection and classification is a key challenge for the protection of High Voltage DC (HVDC) transmission lines. In this paper, the Teager-Kaiser Energy Operator (TKEO) algorithm associated with a decision tree-based fault classi-fier is proposed to detect and classify various DC faults. The Change Identification Filter is applied to the average and differential current components, to detect the first instant of fault occurrence (above threshold) and register a Change Identified Point (CIP). Further, if a CIP is registered for a positive or negative line, only three samples of currents (i.e., CIP and each side of CIP) are sent to this research TKEO algorithm, which produces their respective 8 indices through which the, fault can be detected along with its classification. The new approach enables quicker detection allowing utility grids to be restored as soon as possible. This novel approach also reduces computing complexity and the time required to identify faults with classification. The importance and accuracy of this research scheme are also thor-oughly tested and compared with other methods for various faults on HVDC transmission lines.

Literature Survey 5

Detection and classification of transmission line transient faults based on graph convolutional neural network

Authors:

Published Date: 30-Apr-2021

DOI: 10.17775/CSEEJPES.2020.04970

Literature

This research present a novel transient fault detection and classification approach in power transmission lines based on graph convolutional neural network. Compared with the existing techniques, this research approach considers explicit spatial information in sampling sequences as prior knowledge and it has stronger feature extraction ability. On this basis, a framework for transient fault detection and classification is created. Graph structure is generated to provide topology information to the task. This research approach takes the adjacency matrix of topology graph and the bus voltage signals during a sampling period after transient faults as inputs, and outputs the predicted classification results rapidly. Furthermore, this research approach is tested in various situations and its generalization ability is verified by experimental results. The results show that this research approach can detect and classify transient faults more effectively than the existing techniques, and it is practical for online transmission line protection for its rapidness, high robustness and generalization ability.

Literature Survey 6

Convolutional- and Deep Learning-Based Techniques for Time Series Ordinal Classification

Authors: Rafael Ayllón-Gavilán, David Guijo-Rubio, Pedro Antonio Gutiérrez, Anthony Bagnall, César Hervás-Martínez

Published Date: 27-Nov-2024

DOI: 10.1109/TCYB.2024.3498100

Literature

Time-series classification (TSC) covers the supervised learning problem where input data is provided in the form of series of values observed through repeated measurements over time, and whose objective is to predict the category to which they belong. When the class values are ordinal, classifiers that take this into account can perform better than nominal classifiers. Time-series ordinal classification (TSOC) is the field bridging this gap, yet unexplored in the literature. There are a wide range of time-series problems showing an ordered label structure, and TSC techniques that ignore the order relationship discard useful information. Hence, this article presents the first benchmarking of TSOC methodologies, exploiting the ordering of the target labels to boost the performance of current TSC state of the art. Both convolutional- and deep-learning-based methodologies (among the best performing alternatives for nominal TSC) are adapted for TSOC. For the experiments, a selection of 29 ordinal problems has been made. In this way, this article contributes to the establishment of the state of the art in TSOC. The results obtained by ordinal versions are found to be significantly better than current nominal TSC techniques in terms of ordinal performance metrics, outlining the importance of considering the ordering of the labels when dealing with this kind of problems.

Literature Survey 7

A Phase-Cum-Time Variant Fuzzy Time Series Model for Forecasting Non-Stationary Time Series and Its Application to the Stock Market

Authors: A. J. Saleena, C. Jessy John, G. Rubell Marion Lincy

Published Date: 11-Oct-2024

DOI: 10.1109/ACCESS.2024.3478824

Literature

Non-stationary time series plays a prominent role in the analysis of performance time series of many real-world systems. Recently, fuzzy time series models have been extended to forecast non-stationary time series. Over different phases of time, performance time series may show drastic changes. Therefore, a non-stationary time series is partitioned according to different phases of time. These phases may be taken as weeks, months, or years. Over different phases of time, the universe of discourse, knowledge base, and rule base may vary. A common constraint in the modelling of time-variant fuzzy time series is their incapability to address the phase change of time in the model and accordingly incorporate the necessary changes in the universe of discourse, knowledge base, and rule base over different time phases. To address this issue, a Phase-cum-Time Variant Fuzzy Time Series Model (PTVFTS) is presented in this paper. The PTVFTS model is developed so that it will address the problem of phase change as well as the time variations within each phase simultaneously. The concept of the model rebuilding process is applied to handle changes over each phase and the modified parameter adaptation technique is used for the time variations within each phase. The developed model is applied to the daily closing price time series for the years 2017, 2018, 2019, 2020, 2021, and 2022 separately of stock market indices, NASDAQ, S&P 500, Dow Jones, and TAIEX. The comparison of the developed model is made with the time-variant fuzzy time series model known as the non-stationary fuzzy time series model (NSFTS), Dynamic Evolving Neural-Fuzzy Inference System (DENFIS) model, Long Short-Term Memory (LSTM) model, and the classical Auto-Regressive Integrated Moving Average (ARIMA) model. The efficiency of the developed PTVFTS model is tested using forecasting metrics and statistical tests. The comparison shows that the developed model is more efficient in forecasting phase-cum-time variant non-stationary time series.

Literature Survey 8

STD: A Seasonal-Trend-Dispersion Decomposition of Time Series

Authors: Grzegorz Dudek

Published Date: 19-Apr-2023

DOI: 10.1109/TKDE.2023.3268125

Literature

The decomposition of a time series is an essential task that helps to understand its very nature. It facilitates the analysis and forecasting of complex time series expressing various hidden components such as the trend, seasonal components, cyclic components and irregular fluctuations. Therefore, it is crucial in many fields for forecasting and decision-making processes. In recent years, many methods of time series decomposition have been developed, which extract and reveal different time series properties. Unfortunately, they neglect a very important property, i.e., time series variance. To deal with heteroscedasticity in time series, the method proposed in this work – a seasonal-trend-dispersion decomposition (STD) – extracts the trend, seasonal component and component related to the dispersion of the time series. This research define STD decomposition in two ways: with and without an irregular component. This research show how STD can be used for time series analysis and forecasting.

Literature Survey 9

A Novel Method for IPTV Customer Behavior Analysis Using Time Series

Authors: Tomislav Hlupic, Dražen Oreščanin, Mirta Baranovic

Published Date: 04-Apr-2022

DOI: 10.1109/ACCESS.2022.3164409

Literature

Internet Protocol Television (IPTV) has had a significant impact on live TV content consumption in the past decade, as improvements in the broadband speed have allowed more data volume to be delivered. In addition to existing infrastructure, which is mostly based on the set top boxes, new content providers have emerged, utilizing newly developed proprietary streaming platforms. As the number of IPTV users grew, more volume and variety of data became available for analysis. By analyzing stored user actions, it is possible to create a multivariate time series that represents user behavior over time. The approach presented in the paper is based on multivariate time series generation from user data and determining the similarity between them. Time series are created for each user based on this research quantified action sets, grouped in the feature groups and summarized over time. The action sets and feature groups can be adjusted to a certain IPTV platform. The end result of the analysis is the similarity score matrix, generated by calculating the similarities of all users' time series, where the similarity measure calculation can be chosen arbitrarily.

Literature Survey 10

Extraction of Instantaneous Frequencies and Amplitudes in Nonstationary Time-Series Data

Authors: Daniel E. Shea, Rajiv Giridharagopal, David S. Ginger, Steven L. Brunton, J. Nathan Kutz

Published Date: 08-Jun-2021

DOI: 10.1109/ACCESS.2021.3087595

Literature

Time-series analysis is critical for a diversity of applications in science and engineering. By leveraging the strengths of modern gradient descent algorithms, the Fourier transform, multi-resolution analysis, and Bayesian spectral analysis, this research propose a data-driven approach to time-frequency analysis that circumvents many of the shortcomings of classic approaches, including the extraction of nonstationary signals with discontinuities in their behavior. The method introduced is equivalent to a nonstationary Fourier mode decomposition (NFMD) for nonstationary and nonlinear temporal signals, allowing for the accurate identification of instantaneous frequencies and their amplitudes. The method is demonstrated on a diversity of time-series data, including on data from cantilever-based electrostatic force microscopy to quantify the time-dependent evolution of charging dynamics at the nanoscale.

Problem Statement

Accurate detection and classification of transmission line faults are critical for maintaining power system stability and preventing cascading failures. While **Convolutional Neural Network**-based approaches achieve strong fault detection performance, they struggle with multi-class discrimination due to limited temporal dependency modeling and regression-based ambiguity. Closely related asymmetrical faults exhibit subtle waveform variations that require both localized feature extraction and long-range sequential understanding.

Therefore, there is a need for a hybrid deep learning architecture capable of integrating convolutional feature extraction with sequential memory modeling and adaptive attention weighting. The system must enhance classification separability while maintaining computational feasibility for real-time smart grid deployment.

$$L = \alpha L_{\text{CNN}} + \beta L_{\text{BiLSTM}} + \gamma L_{\text{Attention}}$$

Objectives

- **Enhancing Multi-Class Accuracy:** To improve asymmetrical fault classification performance by integrating convolutional and sequential deep learning mechanisms.
- **Capturing Long-Term Dependencies:** To incorporate **Bidirectional Long Short-Term Memory Network** for modeling forward and backward temporal correlations.
- **Adaptive Temporal Weighting:** To implement **Self-Attention Mechanism** for dynamically emphasizing significant transient intervals.
- **Improving System Generalization:** To design a scalable architecture capable of adapting to larger and more complex bus systems.
- **Ensuring Real-Time Deployment:** To optimize computational efficiency for integration into intelligent protective relay systems.

Motivation

Modern smart grids are becoming increasingly complex due to renewable energy integration and distributed generation. These developments introduce diverse transient disturbances that require advanced intelligent protection strategies. Although **Convolutional Neural Network** models provide accurate detection, their classification performance is limited when handling closely related asymmetrical faults.

Integrating sequential memory architectures and adaptive attention mechanisms can significantly enhance contextual understanding of fault signatures. By combining convolutional feature extraction with bidirectional temporal learning and attention weighting, classification ambiguity can be minimized while maintaining detection precision. This motivates us to do this project.

Proposed System

The proposed ChronoGrid FusionNet Model consists of multi-scale **One-Dimensional Convolutional Neural Network** layers for extracting localized transient features from six-channel voltage and current signals. Kernel sizes of 3 and 5 enable capture of both short-duration spikes and broader waveform distortions.

Extracted features are then processed by a **Bidirectional Long Short-Term Memory Network**, which models forward and backward temporal dependencies. A **Self-Attention Mechanism** assigns adaptive weights to important time steps during transient intervals. Finally, a Softmax classification layer generates multi-class fault predictions. The architecture is optimized using Adaptive Moment Estimation Optimizer and categorical cross-entropy loss function to ensure stable convergence.

Proposed System Advantages

- **Enhanced Temporal Modeling:** Integration of **Bidirectional Long Short-Term Memory Network** improves sequential dependency learning.
- **Attention-Based Feature Prioritization: Self-Attention Mechanism** dynamically emphasizes critical transient intervals.
- **Improved Classification Separability:** Softmax-based output reduces ambiguity compared to regression layers.
- **Higher Robustness:** Hybrid architecture adapts effectively to noisy and complex grid environments.
- **Real-Time Efficiency:** Optimized design maintains computational feasibility for smart grid protection deployment.

Proposed Algorithms

- One-Dimensional Convolutional Neural Network
- Bidirectional Long Short-Term Memory Network
- ChronoGrid FusionNet Hybrid Deep Learning Algorithm

Proposed Algorithm Advantages

- **One-Dimensional Convolutional Neural Network: Advantages**
- Efficient extraction of localized transient waveform features.
- Reduced preprocessing requirement due to direct raw signal processing.
- Lower computational complexity for real-time detection tasks.
- **Bidirectional Long Short-Term Memory Network: Advantages**
- Captures both forward and backward temporal dependencies.
- Improves modeling of sequential voltage-current dynamics.
- Enhances classification stability across varying fault durations.
- **ChronoGrid FusionNet Hybrid Deep Learning Algorithm: Advantages**
- Combines spatial-temporal feature learning in a unified framework.
- Reduces classification ambiguity using attention-guided weighting.
- Improves overall accuracy and robustness compared to standalone convolutional models.

Project Modules

Module 1

Data Acquisition and Preprocessing Module

This module manages voltage and current time-series data collection from simulated transmission systems. It performs normalization, segmentation, labeling, and noise filtering to prepare structured datasets for hybrid deep learning training. Data augmentation strategies are applied to improve generalization across various fault scenarios.

Module 2

Hybrid Deep Learning Model Development Module

This module implements the ChronoGrid FusionNet architecture integrating **One-Dimensional Convolutional Neural Network**, **Bidirectional Long Short-Term Memory Network**, and **Self-Attention Mechanism**. It includes network configuration, optimization strategy design, hyperparameter tuning, and performance evaluation using **Root Mean Square Error** and classification accuracy metrics.

Module 3

Performance Evaluation and Deployment Module

This module evaluates system performance using confusion matrix, precision, recall, and F1-score metrics. Comparative analysis with baseline **Convolutional Neural Network** models is conducted. Deployment feasibility assessment includes computational complexity evaluation and latency analysis for smart grid integration.

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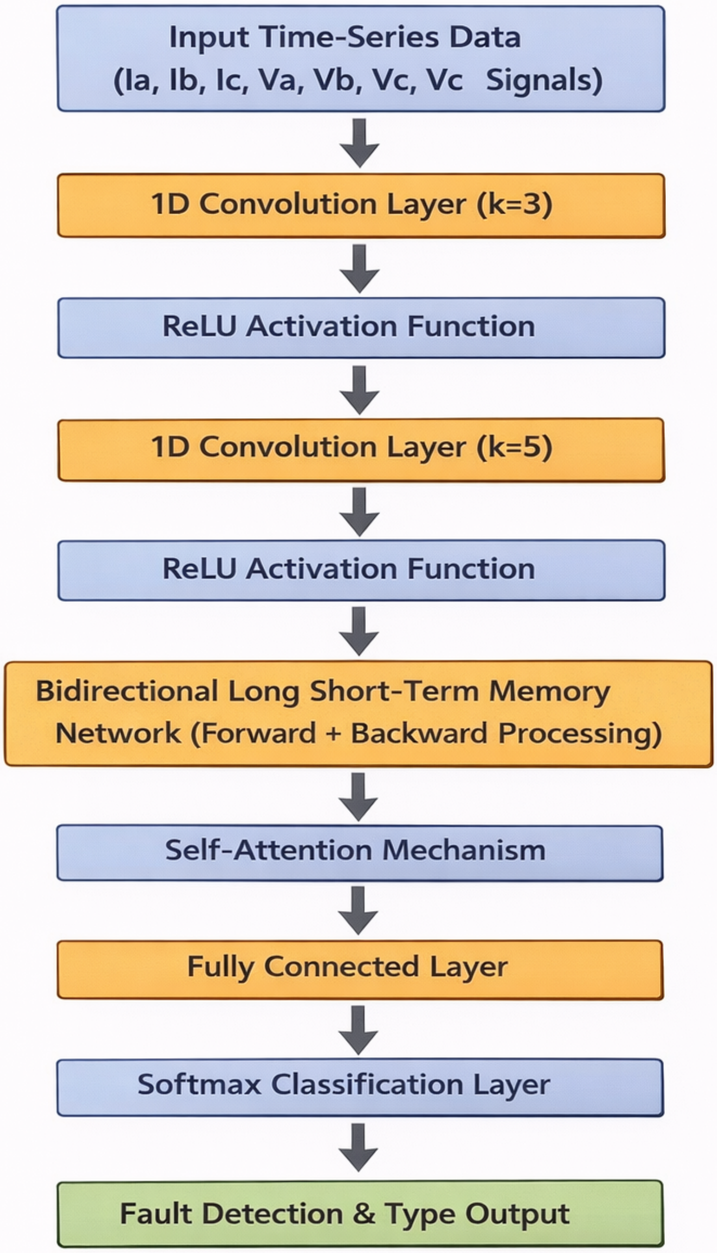
Hardware Requirements

- **Processor:** Minimum i5 Dual Core
- Ethernet connection (LAN) OR a wireless adapter (Wi-Fi)
- **Hard Drive:** Minimum 200 GB; Recommended 200 GB or more
- **Memory (RAM):** Minimum 16 GB; Recommended 32 GB or above
- Networkcard to connect Internet

Software Requirements

- Python For AI/ML/DL Programming
- Jupyter Notebook or PyCharm IDE (Integrated Development Environment) for Development.
- PyTorch or TensorFlow for Deep Learning Coding.
- Sklearn for Machine Learning/Feature Extraction/Evaluation Metrics Coding.
- Numpy for implementing Linear Algebra.
- Plotly for Data Visualization (For Graphs).
- Matplotlib for Data Visualization (For Graphs).
- Seaborn for Data Visualization (For Graphs).
- Pandas for dealing with Tabular Data.
- Cv2 for Image Processing Coding.
- Pillow for Image Processing Coding.

Architecture


Input Time-Series Data
(Ia, Ib, Ic, Va, Vb, Vc, Vc Signals)
1D Convolution Layer (k=3)
ReLU Activation Function
1D Convolution Layer (k=5)
ReLU Activation Function
Bidirectional Long Short-Term Memory Network (Forward + Backward Processing)
Self-Attention Mechanism
Fully Connected Layer
Softmax Classification Layer
Fault Detection & Type Output

Conclusion

The ChronoGrid FusionNet Model significantly enhances transmission line fault detection and classification by integrating convolutional, sequential, and attention-based learning mechanisms. While the baseline **Convolutional Neural Network** provides strong detection accuracy, classification performance is improved through **Bidirectional Long Short-Term Memory Network** and **Self-Attention Mechanism**. The hybrid architecture effectively captures localized transient distortions and long-term temporal dependencies, ensuring superior discrimination among asymmetrical fault types. The proposed framework maintains computational efficiency and scalability for real-time smart grid protection applications.

Future Work

Future research will involve validating the proposed hybrid model using real-world phasor measurement unit datasets and extending evaluation to larger IEEE 39-bus and IEEE 118-bus systems. Integration with transformer-based temporal encoders and graph neural networks will be explored to enhance spatial-temporal modeling across interconnected buses. Model compression techniques such as quantization and knowledge distillation will also be investigated for edge deployment in intelligent protective relays. Cybersecurity robustness analysis will be conducted to ensure secure smart grid operation.

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